

Projektna naloga
Prevajanje programskih jezikov

1. Uvod

Razvili bomo domensko specifični jezik za opis podatkov o nepremičninah.

2. Konstrukti

2.1. Osnovni vrednostni tipi

V jeziku bomo uporabljali naslednje osnovne tipe:

- Realna števila za opis cen, kvadratur, koordinat (npr. 1000, 55.5, -12),
- Nize za shranjevanje imen, lokacij, opisov (npr. "Maribor"),
- Logične vrednosti za pogoje in validacijo (true, false),
- Točke za geografsko dolžino in širino ((15.645501, 46.557596)),
- Enoto kot nevtralni element (nil).

2.2. Strukturni bloki

Korenski element za skupino oglasov je listings.

```
listings "Maribor" {  
  ESTATES  
}
```

Posamezen oglas nepremičnine je estate.

```
estate "Stanovanje Center" {  
  PROPERTIES  
}
```

Meje parcele bodo zapisane s pomočjo parcel, kjer je zadnja točka enaka prvi.

```
parcel {  
  COMMANDS  
}
```

2.3. Lastnosti

Znotraj estate imamo naslednje lastnosti:

LASTNOSTI	PRIMER	TIP
type	type stanovanje;	identifikator (stanovanje/hiša/posest)
offerType	offerType prodaja;	identifikator (prodaja/najem)
location	location (15.645501, 46.557596);	točka
price	price 1000 * 12;	izraz -> število
size	size 100;	izraz -> število
region	region »Podravska«;	niz
neighborhood	neighborhood »Center«;	niz
description	description »Nova hiša«;	niz
source	source »nepremicnine.si«;	niz
set	set(»ogrevanje«, »plin«);	ključ-vrednost

2.4. Geometrijski ukazi

UKAZ	OPIS
line (POINT, POINT)	črta med točkama
box (POINT, POINT)	pravokotnik (zgoraj-levo, spodaj-desno)
bend (POINT, POINT, ANGLE)	krivulja s kotom ukrivljenosti

2.5. Razširjeni konstrukti

Konstantna točka za večkratno uporabo (točka, število, niz) z let:

```
let center = (15.6467, 46.5547);  
let m2Price = 100;
```

Matematični izrazi z operatorji +, -, *, /. Do component točke dostopamo z fst(p), snd(p):

```
price m2Price * size;  
location (fst(center) + 0.01, snd(center));
```

Validacija za price <= 0, size <= 0, nezaprte poligone, prekrivanje parcel.

Komentarji:

```
// Komentar do konca vrstice
```

2.6. Primer

```
// Skupina oglasov za Maribor
listings "Maribor" {

  let center = (15.6467, 46.5547);
  let m2price = 2500;

  estate "Stanovanje Center" {
    type stanovanje;
    offerType prodaja;
    location center;
    size 65;
    price m2price * 65;
    region "Podravska";
    neighborhood "Center";
    description "Stanovanje v centru";
    source "nepremicnine.si";
    set("ogrevanje", "plin");

    parcel {
      line ((15.645, 46.553), (15.648, 46.553));
      line ((15.648, 46.553), (15.648, 46.556));
      bend ((15.648, 46.556), (15.645, 46.556), 20);
      line ((15.645, 46.556), (15.645, 46.553))
    }
  }

  estate "Hiša Tabor" {
    type hiša;
    offerType najem;
    location (fst(center) + 0.01, snd(center) - 0.02);
    size 120;
    price m2price * 120 - 5000;
    region "Podravska";
    neighborhood "Tabor";
    description "Hiša z vrtom";
    source "24nep.si";
    set("zgrajeno", 1998);

    parcel {
      box ((15.652, 46.548), (15.654, 46.546))
    }
  }
}
```

2.7. Definicija

Definicija leksemov

Preskočimo:

presledki $::= [\backslash t \backslash r \backslash n]^+$

komentar $::= "//" [^\backslash n]^*$

Literali:

števila $::= [0-9]^+ ("." [0-9]^+)?$

nizi $::= "'" [^\backslash n \backslash r]^* "'"$

identifikatorji $::= L (L \mid [0-9] \mid "_")^*$

$L = [a-zA-Z_]\mid [\text{À-Öø-öø-ſ}]$

Rezervirane besede:

listings estate parcel let type offerType location price size

region neighborhood description source set

line box bend fst snd true false nil

Operatorji:

lbrace $::= "{"$

rbrace $::= "}"$

lparen $::= "("$

rparen $::= ")"$

semicolon $::= ";"$

comma $::= ","$

assign $::= "="$

plus $::= "+"$

minus $::= "-"$

times $::= "*"$

divide $::= "/"$

Konec datoteke:

eof

2.8. BNF

Program ::= ListingsDecl eof;

ListingsDecl ::= listings string lbrace ListingsBody rbrace;

ListingsBody ::= ListingsStmt ListingsBody | ϵ ;

ListingsStmt ::= LetDecl | EstateDecl;

EstateDecl ::= estate string lbrace EstateBody rbrace;

EstateBody ::= EstateStmt EstateBody | ϵ ;

EstateStmt ::= type identifier semicolon

 | offerType identifier semicolon

 | location PointExpr semicolon

 | price Expr semicolon

 | size Expr semicolon

 | region string semicolon

 | neighborhood string semicolon

 | description string semicolon

 | source string semicolon

 | SetStmt

 | ParcelDecl

 | LetDecl;

LetDecl ::= let identifier assign LetValue semicolon;

LetValue ::= Point | Expr;

SetStmt ::= set lparen string comma SetValue rparen semicolon;

SetValue ::= Point | Expr;

ParcelDecl ::= parcel lbrace ParcelBody rbrace;

ParcelBody ::= GeoCommand GeoCommandsTail | ϵ ;

GeoCommandsTail ::= semicolon GeoCommand GeoCommandsTail

| semicolon | ϵ ;

GeoCommand ::= line lparen PointExpr comma PointExpr rparen

| box lparen PointExpr comma PointExpr rparen

| bend lparen PointExpr comma PointExpr comma Expr rparen;

PointExpr ::= Point | identifier;

Point ::= lparen Expr comma Expr rparen;

Expr ::= Additive;

Additive ::= Multiplicative Additive';

Additive' ::= plus Multiplicative Additive'

| minus Multiplicative Additive'

| ϵ ;

Multiplicative ::= Unary Multiplicative';

Multiplicative' ::= times Unary Multiplicative'

| divide Unary Multiplicative'

| ϵ ;

Unary ::= minus Primary | plus Primary | Primary;

Primary ::= number

| string

| identifier

| true

| false

| nil

| fst lparen Expr rparen

| snd lparen Expr rparen

| lparen Expr rparen;

2.9. FIRST in FOLLOW množice

Začetni simbol: Program. \$ = konec vnosa

FIRST:

Program { listings }

ListingsDecl { listings }

ListingsBody { let, estate, ϵ }

ListingsStmt { let, estate }

EstateDecl { estate }

EstateBody { type, offerType, location, price, size, region, neighborhood,
description, source, set, parcel, let, ϵ }

EstateStmt { type, offerType, location, price, size, region, neighborhood,
description, source, set, parcel, let }

LetDecl { let }

LetValue { lparen, minus, plus, number, string, identifier, true, false, nil, fst, snd }

SetStmt { set }

SetValue { lparen, minus, plus, number, string, identifier, true, false, nil, fst, snd }

ParcelDecl { parcel }

ParcelBody { line, box, bend, ϵ }

GeoCommandsTail { semicolon, ϵ }

GeoCommand { line, box, bend }

PointExpr { lparen, identifier }

Point { lparen }

Expr { minus, plus, number, string, identifier, true, false, nil, fst, snd, lparen }

Additive { minus, plus, number, string, identifier, true, false, nil, fst, snd, lparen }

Additive' { plus, minus, ϵ }

Multiplicative { minus, plus, number, string, identifier, true, false, nil, fst, snd, lparen }

Multiplicative' { times, divide, ϵ }

Unary { minus, plus, number, string, identifier, true, false, nil, fst, snd, lparen }

Primary { number, string, identifier, true, false, nil, fst, snd, lparen }

FOLLOW:

```
Program      { $ }
ListingsDecl { eof }
ListingsBody { rbrace }
ListingsStmt { let, estate, rbrace }
EstateDecl   { let, estate, rbrace }
EstateBody   { rbrace }
EstateStmt   { type, offerType, location, price, size, region, neighborhood,
               description, source, set, parcel, let, rbrace }
LetDecl      { type, offerType, location, price, size, region, neighborhood,
               description, source, set, parcel, let, estate, rbrace }
LetValue     { semicolon }
SetStmt      { type, offerType, location, price, size, region, neighborhood,
               description, source, set, parcel, let, rbrace }
SetValue     { rparen }
ParcelDecl   { type, offerType, location, price, size, region, neighborhood,
               description, source, set, parcel, let, rbrace }
ParcelBody   { rbrace }
GeoCommandsTail { rbrace }
GeoCommand   { semicolon, rbrace }
PointExpr    { semicolon, comma, rparen }
Point        { semicolon, comma, rparen }
Expr         { semicolon, comma, rparen }
Additive     { semicolon, comma, rparen }
Additive'    { semicolon, comma, rparen }
Multiplicative { plus, minus, semicolon, comma, rparen }
Multiplicative' { plus, minus, semicolon, comma, rparen }
Unary        { times, divide, plus, minus, semicolon, comma, rparen }
Primary      { times, divide, plus, minus, semicolon, comma, rparen }
```

2.10. Primeri

```
listings "Maribor" {  
  let center = (15.6467, 46.5547);  
  let m2price = 2500;  
  
  estate "Garsonjera Center" {  
    type stanovanje;  
    location (15.60, 46.55);  
  }  
  
  estate "Stanovanje Center" {  
    type stanovanje;  
    offerType prodaja;  
    location center;  
    size 65;  
    price m2price * 65;  
    region "Podravska";  
    neighborhood "Center";  
    description "Stanovanje v mestnem jedru";  
    source "nepremicnine.si";  
    set("ogrevanje", "plin");  
    parcel {  
      line ((15.640, 46.550), (15.645, 46.550));  
      line ((15.645, 46.550), (15.645, 46.554));  
      bend ((15.645, 46.554), (15.640, 46.554), 20);  
      line ((15.640, 46.554), (15.640, 46.550))  
    }  
  }  
  
  estate "Hiša Tabor" {  
    type hiša;  
    offerType najem;  
    location (fst(center) + 0.01, snd(center) - 0.01);
```

```
size 120;
price m2price * 120 - 5000;
region "Podravska";
neighborhood "Tabor";
set("zgrajeno", 1998);
parcel {
    box ((15.700, 46.556), (15.706, 46.550))
}
}
```

```
estate "Parcela Pobrežje" {
    type posest;
    location (15.752, 46.553);
    parcel {
        line ((15.750, 46.550), (15.755, 46.550));
        line ((15.755, 46.550), (15.752, 46.555));
        line ((15.752, 46.555), (15.750, 46.550))
    }
}
```

```
estate "Stanovanje Studenci" {
    let area = 80;
    let unit = 1800;
    type stanovanje;
    offerType prodaja;
    location (15.61, 46.56);
    size area;
    price unit * area + 5000 - 2000;
}
```

```
estate "Zemljišče Južni rob" {
    type posest;
    location (-15.123, -46.456);
    description "Preizkus negativnih koordinat";
}
```

```
}
```

```
estate "Hiša ob Dravi" {  
  type hiša;  
  location (15.62, 46.57);  
  region "Štajerska";  
  neighborhood "Pobrezje";  
  description "Hiša ob reki Dravi";  
}
```

```
estate "Vila Kalvarija" {  
  type stanovanje;  
  location (15.63, 46.58);  
  set("ogrevanje", "toplotna črpalka");  
  set("nadstropja", 2);  
  set("cena_na_kvadrat", 3.14);  
  set("oddano", true);  
  set("upravnik", nil);  
  set("vokal", (15.6, 46.5));  
}
```

```
estate "Posest Pekre" {  
  type hiša;  
  location (15.64, 46.59);  
  size  $10 + 2 * 3$ ;  
  price  $(100 + 50) * 2 - 10 / 5$ ;  
}
```

```
estate "Neveljaven oglas" {  
  type posest;  
  offerType najem;  
  location (15.80, 46.55);  
  size -5;  
  price 0;
```

```
parcel {  
  line ((15.800, 46.550), (15.805, 46.550));  
  line ((15.805, 46.550), (15.805, 46.555))  
}  
}  
}
```